Example 7.17

A person bought 12% stock for ₹54,000 at a discount of 17%. If he paid 1% brokerage, find the percentage of his income.

Solution:

Face value = ₹100

Market value = ₹(100 – 17 + 1) = ₹84

$\therefore$ percentage of his income = $\frac{(12\times100)}{84}$

$= \frac{100}{7} = 14\frac{2}{7}$

$\therefore$ % of income = $14\frac{2}{7}$ %

Example 7.18

Equal amounts are invested in 10% stock at 89 and 7% stock at 90 (1% brokerage paid in both transactions). If 10% stock bought ₹100 more by way of dividend income than the other, find the amount invested in each stock.

Solution:

Let x be the amount invested in each stock

Income on 10% stock at 89:

F.V. = ₹100, M.V. = ₹90

Number of shares = $\frac{investments}{M.V.} = \frac{x}{90}$

Annual income = $\frac{x}{90}\times\frac{10}{100}\times100$

$= \frac{x}{9}$ ... (1)

Income on 7% stock at 90:

M.V. = ₹91

Number of shares = $\frac{\text{investments}}{MV} = \frac{x}{91}$

Annual income = $\frac{x}{91}\times\frac{7}{100}\times100$

$= \frac{x}{13}$ ... (2)

Given : $\frac{x}{9}-\frac{x}{13} = 100 \Rightarrow x = ₹\ 2925$

$\therefore$ The amount invested in each stock $\approx$ ₹2,925.

Example 7.19

A capital of a company is made up of 1,00,000 preference shares with a dividend rate of 16% and 50,000 ordinary shares.The par value of each of preference and ordinary shares is ₹10.The total profit of a company is ₹ 3,20,000.If ₹40,000 were kept in reserve and ₹20,000 were kept in depreciation fund, what percent of dividend is paid to the ordinary share holders.

Solution:

F.V. = ₹10

Total face value of preference shares

= ₹1,00,000 × 10 = ₹10,00,000

Total face value of ordinary shares

= ₹50,000 × 10 = ₹5,00,000

Total dividend amount paid to shareholders

= ₹(3,20,000 – 40,000 – 20,000) = ₹2,60,000

Dividend for preference shares

$= \frac{16}{100}\times10,00,000 = ₹1,60,000$

Dividend to ordinary shares

= 2,60,000 – 1,60,000 = ₹1,00,000

Dividend rate for ordinary share

$= \frac{Income}{Investment}\times100\%$

Dividend % = $\frac{1,00,000}{5,00,000}\times100 = 20\%$

Example 7.20

A person sells a 20% stocks of face value ₹10,000 at a premium of 42%. With the money obtained he buys a 15% stock at a discount of 22%. What is the change in his income if the brokerage paid is 2%.

Solution:

Step 1: For 20% stocks:

F.V. = ₹100

Income = $\frac{20}{100}\times10000$

= ₹ 2,000 ... (1)

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